

Qi Wu

Optical Metrology · Surface Science · Instrumentation

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Profile. Researcher and hands-on engineer with four years at imec (KU Leuven) working on the optical characterization of liquids and surfaces, and on building the instruments to measure them. Work spans surface modification and wetting, custom ATR-FTIR and contact-angle metrology, optical simulation, and data analysis in Python and MATLAB. Cross-disciplinary background across optics, electrical engineering, and MEMS.

EDUCATION

Ph.D. in Chemistry	KU Leuven & imec, Belgium 2022–2026 (expected)
M.Sc. in Electrical Engineering	University of Twente, the Netherlands 2019–2021
B.Eng. in Electrical Engineering	Northwestern Polytechnical University, China 2015–2019

TECHNICAL SKILLS

Characterization & Metrology: FTIR spectroscopy; ellipsometry; wettability analysis; SEM; AFM.

Instrumentation & Prototyping: CAD design and build of custom metrology systems; re-engineering of commercial instruments; embedded control / microcontrollers; 3D-printing prototyping; measurement automation.

Optical Simulation: FDTD simulation of evanescent-field and near-field optics; spectral correction, processing, and analysis.

Programming & Data: Python and MATLAB for data processing, analysis, and hardware-software integration.

Experiment & Process: design of experiments (DOE); self-assembled monolayers (SAMs) LPCVD; wet-chemical surface processing; 2,000+ hrs cleanroom.

Languages: English (professional), Dutch (beginner), Mandarin Chinese (native).

RESEARCH EXPERIENCE

PhD Researcher *KU Leuven & imec, Leuven, Belgium | 2022–Present*

Superhydrophobic Surface Fabrication by Silane SAM Deposition

- **Surface engineering:** engineered superhydrophobic silicon surfaces via optimized silane self-assembled monolayer (SAM) deposition on patterned substrates, reaching intrinsic and apparent water contact angles above 110° and 140°.
- **Mechanistic analysis:** resolved the deposition mechanism by GATR-FTIR spectroscopy, profiling the kinetic balance between surface –OH loss during HF pre-cleaning and re-oxidation during CVD.
- **Process control:** established a robust pre-cleaning and deposition SOP with >90% run-to-run reproducibility; recognized with the Best Student Paper Award at UCPSS 2025.

Quantitative Optical Metrology of Wetting Defects in HAR Nanopatterns

- **Instrument development:** built a customized ATR module for a commercial FTIR platform with an adjustable optical path, tuning the number of IR reflections to raise signal-to-noise ratio while avoiding absorption saturation.
- **Defect quantification:** quantified nanoscale air-trapping in high-aspect-ratio gaps by solution-phase IR absorption, exploiting the quantitative drop in solution absorption when liquid fails to fill the structure.
- **Optical simulation:** modelled the near-field electromagnetic distribution at the structured liquid–solid interface under total internal reflection (FDTD), linking sub-wavelength geometry to the evanescent field and IR response to calibrate the measured spectra.
- **Method validation:** defined the metrological limits of the technique, establishing the valid geometric window (pattern height < 500 nm, surface fraction < 0.7) for accurate defect quantification on single-material nanopatterns.

Synchronized Cross-View Imaging Platform for Anisotropic Wetting Characterization

- **Platform design:** designed and built a cross-view optical platform for simultaneous, dual-directional imaging of sessile droplets on nano-trenches.
- **Measurement accuracy:** eliminated the timing mismatch of sequential imaging, enabling synchronized capture of anisotropic droplet elongation with improved morphological accuracy.

- **New defect indicator:** identified the structural-parameter regime where conventional wetting models break down, and established real-time anisotropic elongation as a new indicator of nanoscale wetting defects.

Graduate Intern

ARCNL, Amsterdam, the Netherlands | 2021

- **Mechanical simulation:** ran stress-strain simulations (COMSOL) of elastomer microstructure arrays for low-wear wafer carriers.
- **Industrial collaboration:** worked with cleanroom specialists to improve micro-component durability in a research-to-manufacturing setting.

PUBLICATIONS & PRESENTATIONS

Publications

- Q. Wu et al., *"Modulating Silane Monolayer Grafting on Nanopatterned Silicon: Insights from Infrared-Spectroscopy-Based Air-Trapping Monitoring,"* Solid State Phenomena (UCPSS 2025, 17th International Symposium on Ultra Clean Processing of Semiconductor Surfaces), accepted / in press. **Best Student Paper Award.**

Presentations

- Q. Wu, *"Impact of HF Pretreatment on Silane Self-Assembled Monolayers on Silicon Surfaces,"* oral presentation, 18th Mini-Symposium on Liquids, Okayama University (online).
- Q. Wu, *"Optical Method to Characterize the Wetting States on Nano-patterned Surfaces,"* poster presentation, SPP 2171 Summer School.

INTERESTS

Tennis, bouldering, hiking, DIY electronics.